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01 - VHDL Exercise

Hardwaredesign PR - Exercise 01

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Contents

1. Combinatorial Circuit

Combinatorial Circuit

Write an entity and architecture for the system described in Figure 1. This entity has two inputs (`operand_a_i` and `operand_b_i`) and two outputs. The first output `and_o` should be calculated as the logical AND operation of the two inputs and the second output `or_o` should be calculated as the logical OR operation of the two inputs.

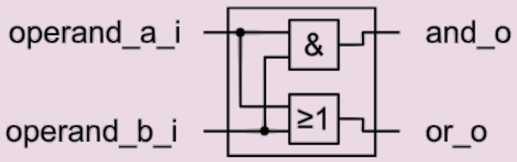


Figure 1: Combinatorial Circuit

To-Do-List

- Write the entity and architecture of the system in VHDL. Use one .vhd file for the entity and one .vhd file for the architecture.
- Write a testbench in VHDL (extra file). Simulate the system using Modelsim and put the screenshots showing the output of the waveform window in the report pdf.
- Discuss the results of the simulation in a few sentences.

1.1 Model Implementation

Code can be found in the attachments.

1.2 Simulation

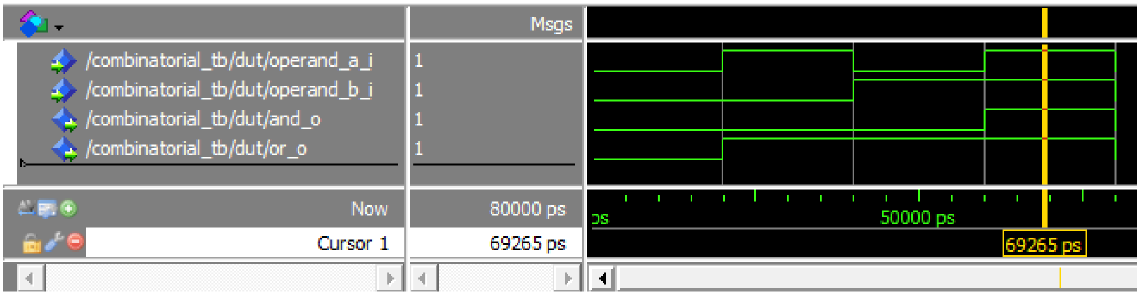


Figure 2: Waveforms for combinatorial circuit

Result

The truthtable of an AND and an OR can be recognized in the Waveform:

a_i	b_i	and_o	or_o
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

2. Adder

Adder

Write an entity and architecture for the system described in Figure 3. This entity has two inputs (`operand_a_i` and `operand_b_i`) and one output. The inputs and outputs of this entity are no single bits but bit vectors (use `std_ulogic` vectors). The output `result_o` should be calculated as the arithmetical sum of the two inputs. The bit width of the inputs (both should have the same bit width) should be defined by a generic `BITWIDTH`. The output bit width should also be defined via this generic (but is of course not equal to this generic...).

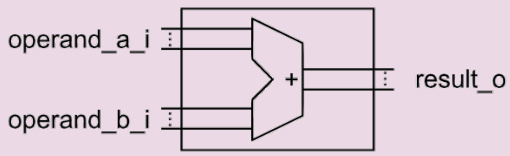


Figure 3: Adder

To-Do-List

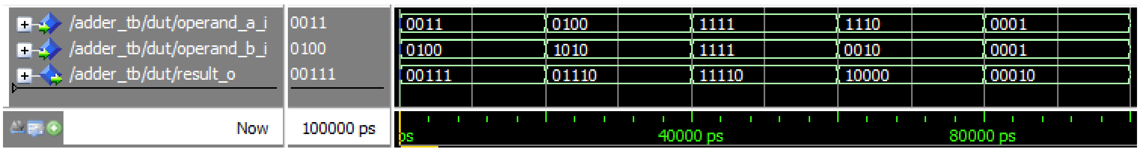
- Write the entity and the architecture of the system in VHDL. Use one .vhd file for the entity and one .vhd file for the architecture. Describe: how you have defined the output bit width? Give the reasons (few sentences) for your definition.
- Write a testbench in VHDL (extra file). Test the system and hand in screenshots showing the output of the waveform window. Do two tests with different bit widths ($\leq$ 3 bits). Think of useful test cases (at least 5 different ones for every bit width). In a few sentences summarize the reasons why you have chosen these test cases. In addition do one test where you add two times the maximum value of the inputs. Discuss the result of the simulation in a few sentences. Do you need additional output bits?

2.1 Model Implementation

Code can be found in the attachments.

2.2 Simulation

4-Bit Adder



12-Bit Adder

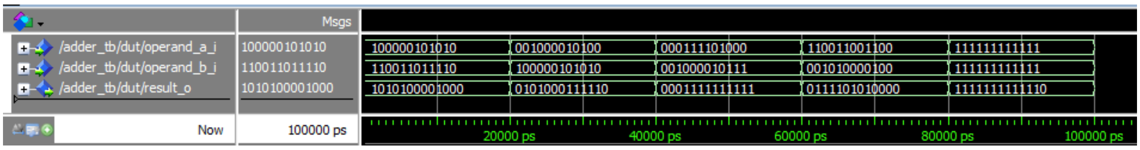


Figure 4: Caption

Die Bitbreite der operanden ist `BITWIDTH-1 downto 0` sodass die Breite inklusive der null genau der Bitbreite entspricht. Im Fall des 4-Bit Adder also von 3 bis 0. Beim Ausgang soll das Carry-Bit berück-sichtigt werden, damit die Zahl dargestellt werden kann wenn das Ergebnis die Bitbreite überschreitet. Die größe des resultates ist beim Addierer maximal 1 Bit größer als der größte Eingangswert. Die dimension des resultates ist daher `BITWIDTH downto 0`.

Als Testwerte wurden Zahlen verwendet, die unterschiedliche Auswirkungen auf das Carry-Bit haben. Bei der 4-Bit Adder Simulation ist gut zu sehen, wie für beide operanden der höchste 4-Bit Wert 15 gewählt wurde. Wie man sieht, bleibt das resultat innerhalb des definierten Bit-Bereiches